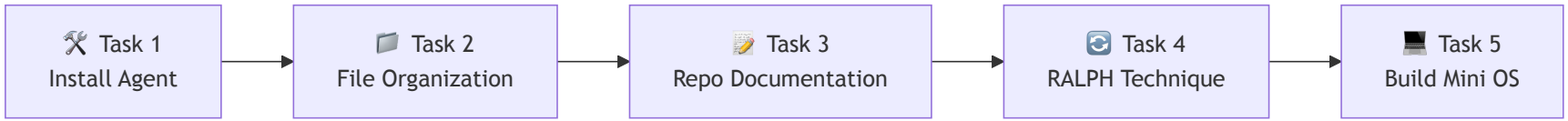
# Operating Systems Lab

## Week 1 – Coding Agents

Korea University Sejong Campus, Department of Computer Science

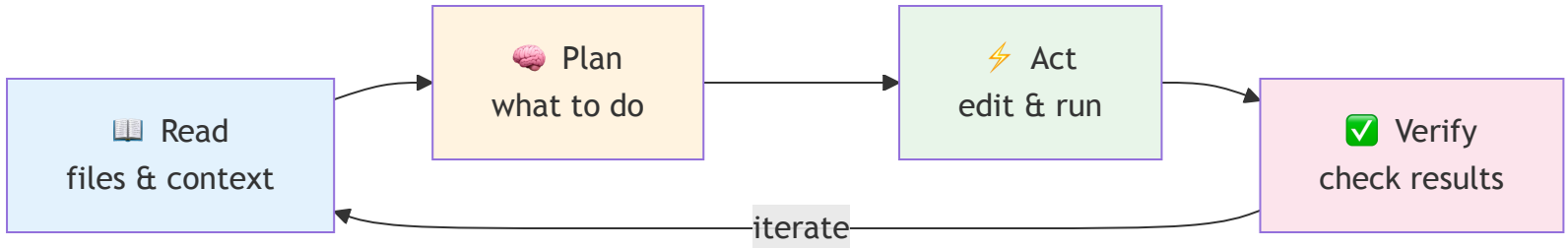
# Lab Overview

- **Goal**: Get comfortable using AI-powered coding agents
- **Duration**: ~50 minutes
- **Submission**: None – exploration lab
- **5 Tasks**: Install → File Org → Repo Docs → RALPH → Mini OS



# What Are Coding Agents?

AI-powered CLI tools that understand and generate code **in context**




- Can read your file system, run commands, and make edits autonomously
- Useful for: scaffolding, refactoring, documentation, debugging

> **Example**: *"Create a Makefile for a C project with debug and release targets"* Agent: reads directory → writes Makefile → confirms build succeeds

# Available Agents

### Gemini CLI

Free

Google's agent

Good default choice

### Claude Code

Paid

Anthropic's agent

Strong multi-file reasoning

### Codex CLI

Paid

OpenAI's agent

Open-source CLI

> **Recommendation**: Install **Gemini CLI** if you have no preference – it's free and easy to set up.

# Task 1 & 2: Install & File Organization

## Task 1 – Install a Coding Agent

1. Follow the official install guide
2. Verify: `gemini --version`
3. Authenticate with your account

## Task 2 – File Organization

Point the agent at a messy directory:

```
"Organize the files in ./downloads
 into subfolders by type and date"
```

Review – did the agent make sensible decisions?

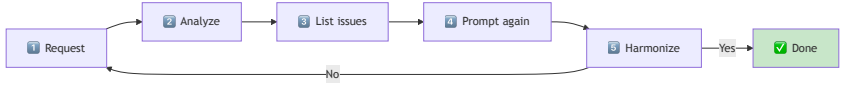
# Task 3 & 4: Repo Docs & RALPH

## Task 3 – GitHub Repo Documentation

- Open an existing GitHub repository
- Ask the agent to generate a `README.md`
- Check: does it accurately describe the project?

## Task 4 – RALPH Technique

**R**equest → **A**nalyze → **L**ist issues → **P**rompt again → **H**armonize

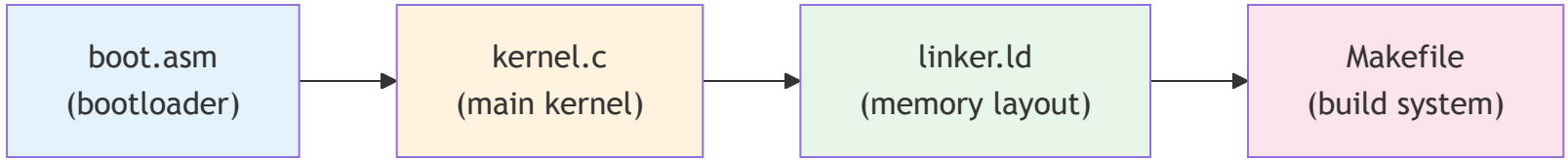



Practice this loop on the README from Task 3.

# Task 5: Build a Mini OS

Ask your coding agent to scaffold a minimal OS kernel in C:

```
"Create a minimal x86 OS kernel in C that boots with GRUB,
 prints 'Hello, OS!' to the screen, and halts.
 Include a Makefile and linker script."
```




- Observe how the agent breaks the problem into files
- You do **not** need to successfully boot it – the **process** matters

# Summary & Next Steps

## What we practiced today

| Task | Skill |
| --- | --- |
| Install agent | Tool setup & authentication |
| File organization | Delegating tasks to AI |
| Repo documentation | Evaluating AI-generated output |
| RALPH technique | Iterative refinement loop |
| Mini OS | Tackling complex systems projects with AI |

**Coming up – Week 2 Lab**: Process system calls ( `fork` , `exec` , `wait` , `pipe` )

> Coding agents are tools – understanding what they produce is still **your** responsibility.